Datasets for Skin Cancer Detection Using Deep Learning

A Verified Survey of Public Datasets and a Recommended Combination

Project: Skin Cancer Detection Using Deep Learning

Companion document to the IEEE paper

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Abstract

This document surveys the public image datasets that are suitable for a deep-learning skin cancer detection project. For each dataset it reports the number of images, imaging modality, diagnostic classes, available metadata, class balance, licence, and download location. Image counts were checked against the official label files or data papers. The document then recommends a combination of six datasets that covers melanoma, basal cell carcinoma (BCC), and squamous cell carcinoma (SCC), provides external and skin-tone-diverse test sets, and avoids the duplicate images that inflate published results.

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1 Why Dataset Choice Matters

A skin cancer classifier can only be as good as the data it learns from. Three properties of public dermatology data strongly affect the results a project reports:

1. **Class imbalance.** Malignant lesions are rare. They make up 0.1% of SLICE-3D, 1.8% of ISIC 2020, and about 20% of HAM10000 (Fig. 2). A model that always predicts "benign" can still look accurate.
2. **Duplicates and overlap.** Many lesions are photographed several times, and the ISIC collections re-use earlier data (e.g., ISIC 2019 contains HAM10000 and BCN20000). Removing duplicates reduced the best ISIC 2020 test AUC of 19 CNNs to 0.80 [1].
3. **Limited diversity.** Across 21 open datasets, skin type was recorded for only 2.1% of images and ethnicity for 1.3%, and most images come from Europe, North America, and Oceania [2]. Models trained on such data perform worse on darker skin [3].

Key idea: use one large dataset for training, and *different* datasets (another hospital, another camera, other skin tones) for testing. A model that is good only on its own training distribution is not ready for patients.

2 Summary of Public Datasets

Table 1 lists the datasets reviewed. Malignant classes are printed in bold. MEL = melanoma, BCC = basal cell carcinoma, SCC = squamous cell carcinoma, AK/AKIEC = actinic keratosis / intra-epithelial carcinoma, NV = nevus, BKL = benign keratosis, DF = dermatofibroma, VASC = vascular lesion.

Table 1: Public skin-cancer datasets (counts from official label files or data papers).

Dataset	Year	Images	Modality	Classes	Metadata
HAM10000 [4]	2018	10,015	Dermoscopic	7: **MEL**, **BCC**, **AKIEC**, NV, BKL, DF, VASC	Age, sex, site, diagnosis method, lesion ID
ISIC 2016 [5]	2016	1,279	Dermoscopic	Binary: **malignant** vs. benign	–
ISIC 2017 [6]	2017	2,750	Dermoscopic	3: **MEL**, seborrheic keratosis, NV	Approx. age, sex
ISIC 2018 [7]	2018	2,594 (T1 train)	Dermoscopic	Task 1 masks; Task 3 = HAM10000 classes	As HAM10000
ISIC 2019 [8]	2019	25,331	Dermoscopic	8: **MEL**, **BCC**, **SCC**, **AK**, NV, BKL, DF, VASC	Age, sex, site, lesion ID
BCN20000 [9]	2024	18,946	Dermoscopic	8 (as ISIC 2019) + out-of-distribution	Age, sex, site, lesion ID
ISIC 2020 [10]	2020	33,126	Dermoscopic	Binary: **MEL** vs. benign	Patient ID, age, sex, site
SLICE-3D (ISIC 2024) [11]	2024	401,059	3D total-body photo crops	Binary: **malignant** (MEL, BCC, SCC) vs. benign	Age, sex, site, lighting, lesion features
MILK10k (ISIC 2025) [12]	2026	10,480	Paired clinical + dermoscopic	11 incl. **BCC**, **SCC/KA**, **MEL**, **AKIEC**	Age, sex, site, skin tone
PH2 [13]	2013	200	Dermoscopic	3: **MEL**, atypical NV, common NV	Masks, dermoscopic criteria
Derm7pt [14]	2019	1,011 cases	Clinical + dermoscopic	**MEL**, **BCC**, NV, SK, misc.	7-point checklist, sex, site
PAD-UFES-20 [15]	2020	2,298	Smartphone	6: **BCC**, **SCC**, **MEL**, ACK, SEK, NEV	21 fields incl. Fitzpatrick
Fitzpatrick 17k [16]	2021	16,577	Clinical atlas	114 conditions; **malignant** / benign / non-neoplastic	Fitzpatrick I–VI
DDI [3]	2022	656	Clinical	Binary: **malignant** vs. benign (biopsy-proven)	Fitzpatrick groups
MED-NODE [17]	2015	170	Clinical	**MEL** vs. NV	–
Dermofit [18]	2013	1,300	Clinical	10 incl. **MEL**, **BCC**, **SCC**, **IEC**	– (paid licence)
SCIN [19]	2024	10,408	Self-taken smartphone	Mostly inflammatory / infectious conditions	Self-reported demographics, skin tone

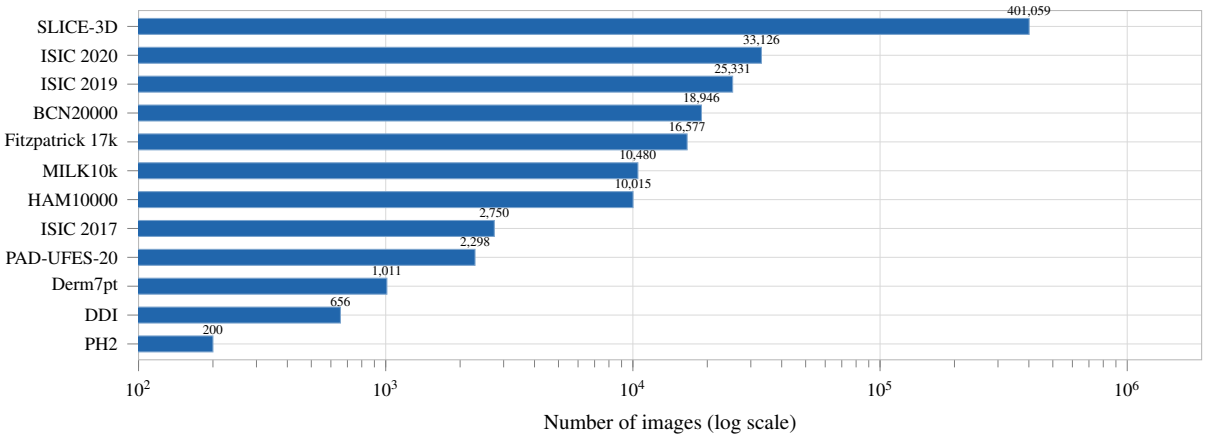


Figure 1: Dataset sizes (log scale). Derm7pt is counted in cases (each has clinical and dermoscopic images).

3 Class Balance and Malignant Prevalence

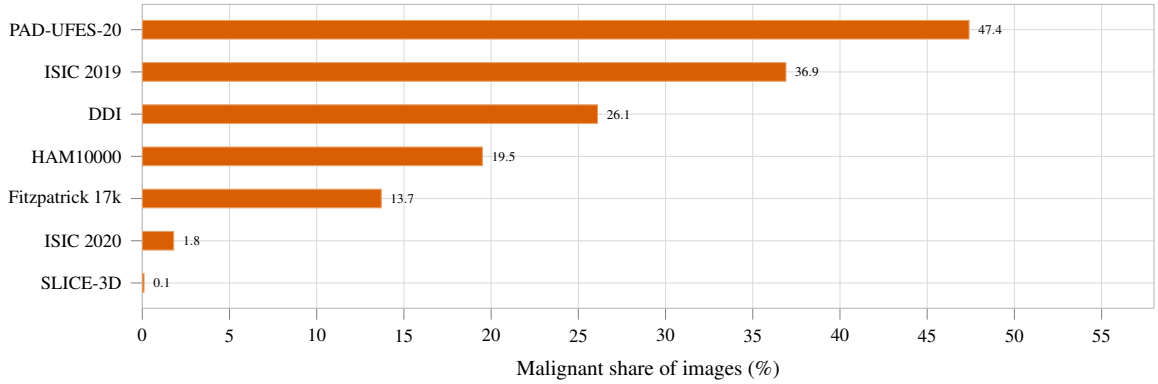


Figure 2: Malignant share per dataset, computed from the official label files. HAM10000 and ISIC 2019 include the AKIEC/AK class. PAD-UFES-20 counts BCC, SCC, and MEL.

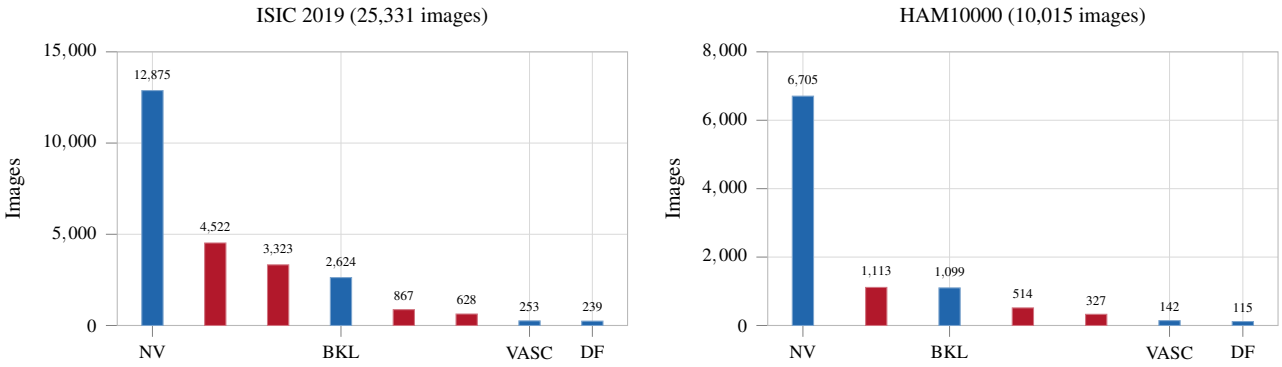


Figure 3: Class distributions of ISIC 2019 and HAM10000 (red = malignant or pre-malignant). In ISIC 2019 the largest class is 54× the smallest.

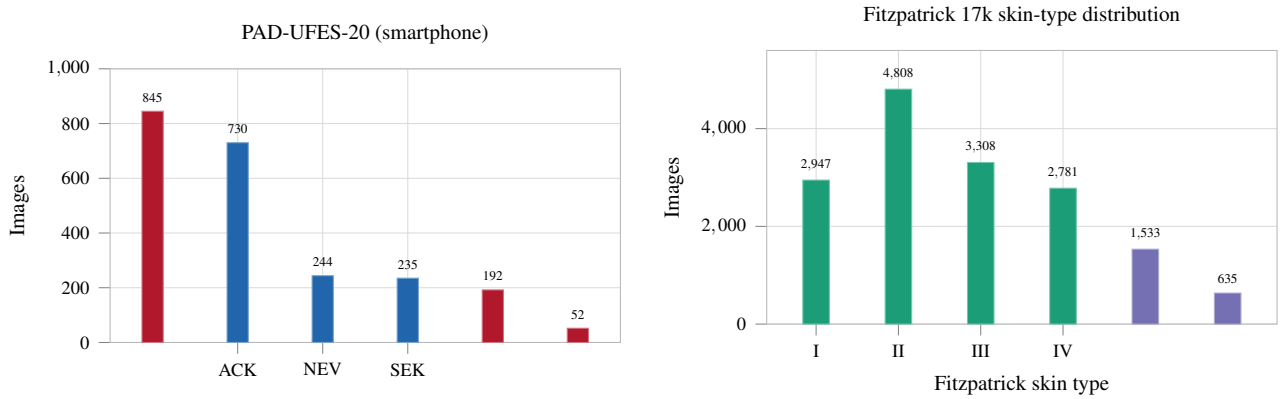


Figure 4: Left: PAD-UFES-20 classes (red = cancers; ACK is pre-malignant). Right: Fitzpatrick 17k skin types. Darker types V–VI (purple) make up only 13% of images.

4 Dataset Profiles

4.1 HAM10000

10,015 dermoscopic images, 7 classes [4]. Collected in Vienna and Queensland. More than half of the diagnoses are confirmed by histopathology (5,340); the rest are confirmed by follow-up, expert consensus, or confocal microscopy. It is the most widely used benchmark and is identical to the ISIC 2018 Task 3 training set. **Caution:** it has 7,470 unique lesions, so several images share a lesion. Split by `lesion_id`, not by image, to avoid leakage. Licence: CC BY-NC 4.0.

4.2 ISIC 2016, 2017, and 2018 Challenges

These early ISIC challenge sets [5–7] are small and dermoscopic. Their main value today is the **lesion segmentation masks**: ISIC 2018 Task 1 provides 2,594 training, 100 validation, and 1,000 test images with expert masks. ISIC 2017 adds a three-class task (melanoma, seborrheic keratosis, nevus). Note: for ISIC 2016 the official label file contains 173 malignant of 900 training images (19.2%), which differs from the figure printed in the challenge paper; cite the label-file count.

4.3 ISIC 2019 (Recommended Main Training Set)

25,331 training images, 8 classes [8]. It combines HAM10000, BCN20000, and MSK images, so it already includes all the major dermoscopic sources and all three skin cancers (MEL 4,522; BCC 3,323; SCC 628; AK 867). Its official metric is balanced accuracy, and its hidden test set (8,238 images) includes an "unknown" class. **Do not add HAM10000 or BCN20000 again**, because they are already inside. Licence: CC BY-NC.

4.4 BCN20000

18,946 images of 5,583 lesions from Hospital Clínic Barcelona [9, 20]. The images are "in the wild": nails, mucosa, and large or hypopigmented lesions make it harder than HAM10000. The earlier arXiv version reports 19,424 images. Licence: CC BY 4.0.

4.5 ISIC 2020 (SIIM-ISIC)

33,126 images from 2,056 patients [10]. Binary melanoma detection with only 584 melanomas (1.8%), close to real-world prevalence. Its patient-level structure (about 16 lesions per patient) enables "ugly duckling" models that compare a lesion with the patient's other lesions. Licence: CC BY-NC 4.0.

4.6 SLICE-3D (ISIC 2024)

401,059 lesion crops from 1,042 patients [11]. The crops come from 3D total-body photographs and have smartphone-like quality. Only 393 lesions are malignant (0.1%), all confirmed by histopathology. The dataset was the basis of the ISIC 2024 triage challenge [21]. Best for triage and extreme-imbalance experiments. A CC BY "permissive" subset of 217,477 images is available.

4.7 MILK10k (ISIC 2025)

5,240 lesions, each with a clinical close-up and a dermoscopic image (10,480 images), 11 classes [12]. 95.7% of lesions were confirmed by biopsy or excision, and each record includes a skin-tone field. The largest class is BCC (48.1% of lesions), so it tests non-melanoma cancer well. **Caution:** some dermoscopic images already exist in the ISIC archive. Remove any `isic_id` that also appears in your training data before testing.

4.8 PAD-UFES-20

2,298 smartphone images of 1,641 lesions from 1,373 patients (Brazil) [15]. All cancers are biopsy-proven. It has up to 21 clinical features, including Fitzpatrick type, lesion diameter, and whether the lesion itches, bleeds, or has grown. This makes it the best test of whether a dermoscopy-trained model works on ordinary phone photos. Licence: CC BY 4.0.

4.9 Fitzpatrick 17k and DDI (Fairness)

Fitzpatrick 17k [16] has 16,577 clinical atlas images with a Fitzpatrick label for each image, but only 13% are types V–VI (Fig. 4). DDI [3] is smaller (656 images, 570 patients) but every diagnosis is biopsy-proven, and it is balanced across skin-tone groups: types I–II 208 images, III–IV 241, and V–VI 207. Models lose substantial performance on DDI, which makes it the standard fairness test.

4.10 PH2 and Derm7pt

PH2 [13] (200 images: 80 common nevi, 80 atypical nevi, 40 melanomas) includes expert masks and dermoscopic criteria and is a good small external test for segmentation. Derm7pt [14] (1,011 cases) annotates the seven-point checklist criteria, which is useful for evaluating concept-based explanations.

4.11 Other Datasets

MED-NODE [17] (170 clinical images) is a small melanoma test set. Dermofit [18] (1,300 images) requires a paid licence. SCIN [19] (10,408 self-taken photos) contains mainly inflammatory and infectious conditions, so it is suited only to out-of-distribution ("not a cancer") testing.

5 Recommended Dataset Combination

Table 2 gives the recommended combination for this project, and Fig. 5 shows how the datasets are used.

Table 2: Recommended datasets and their roles.

Dataset	Role	Why it was chosen
ISIC 2019	Main training (8 classes)	Largest multi-class dermoscopic set; covers MEL, BCC, SCC, and AK; includes HAM10000 and BCN20000
ISIC 2020	Melanoma training	Realistic 1.8% melanoma prevalence; patient-level metadata
ISIC 2018 Task 1	Segmentation training	2,594 expert lesion masks
MILK10k	External test	Newest ISIC benchmark; 11 classes; biopsy-confirmed; clinical + dermoscopic
PAD-UFES-20	External test (smartphone)	Tests transfer from dermoscopy to phone photos; biopsy-proven cancers
DDI	Fairness test	Balanced across skin tones; biopsy-proven
PH2	Segmentation test	Small external set with expert masks

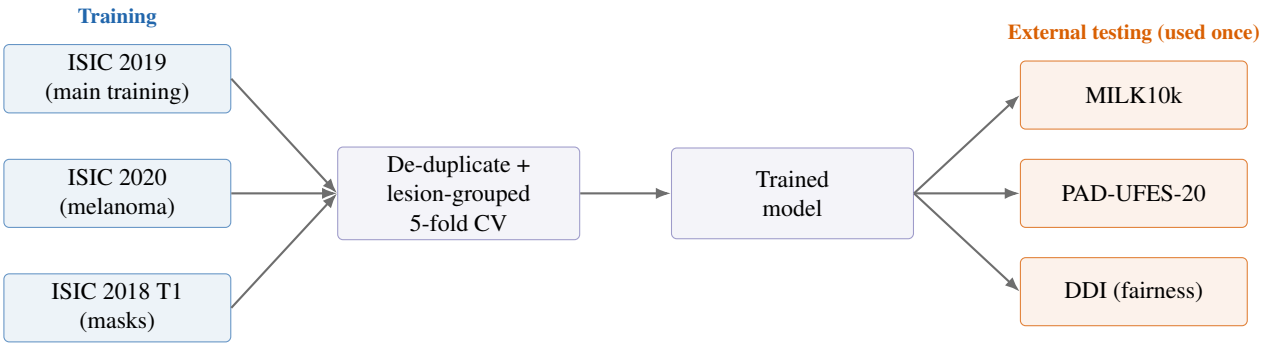


Figure 5: How the recommended datasets are used: training data are cleaned and split by lesion; external sets are used only once, for the final evaluation.

6 Download Links and Licences

Table 3: Where to obtain each recommended dataset.

Dataset	Licence	Access
ISIC 2016–2019	CC-0 / CC BY-NC	https://challenge.isic-archive.com/data/
HAM10000	CC BY-NC 4.0	https://doi.org/10.7910/DVN/DBW86T
ISIC 2020	CC BY-NC 4.0	https://doi.org/10.34970/2020-ds01 (also on Kaggle)
SLICE-3D	CC BY-NC (CC BY subset)	https://doi.org/10.34970/2024-slice-3d
MILK10k	CC BY-NC	https://doi.org/10.34970/648456
BCN20000	CC BY 4.0	https://doi.org/10.6084/m9.figshare.24140028
PAD-UFES-20	CC BY 4.0	https://doi.org/10.17632/zr7vgbcyr2.1
DDI	Research-use agreement	Stanford AIMI shared datasets portal
Fitzpatrick 17k	CC BY-NC-SA 3.0	https://github.com/mattgroh/fitzpatrick17k
PH2	Research use	https://www.fc.up.pt/addi/ph2%20database.html

All recommended licences allow academic research. Most forbid commercial use; for a commercial product, only the CC BY sets (BCN20000, PAD-UFES-20, SLICE-3D permissive subset) are usable without further agreements.

7 Pitfalls and Good Practice

1. **Remove duplicates across datasets** using a perceptual image hash before any split [1]. ISIC 2018 Task 3 = HAM10000; ISIC 2019 ⊃ HAM10000 + BCN20000; MILK10k shares some ISIC images. Whether ISIC 2019 and ISIC 2020 overlap could not be confirmed, so check this with a hash as well.
2. **Split by lesion or patient**, never by image (`GroupKFold` on `lesion_id` or `patient_id`).
3. **Report balanced accuracy and per-class sensitivity**, not only accuracy, because prevalence varies from 0.1% to 47%.
4. **Report results per skin type** on DDI and PAD-UFES-20.
5. **Keep external test sets untouched** until the final evaluation. Tuning on them is a form of leakage.
6. **Harmonize labels** when combining datasets: map AKIEC/AK/ACK to one class and SCC/SCCKA to one class, and keep a written mapping table in the project repository.

Summary: train on ISIC 2019 + ISIC 2020 (segmentation on ISIC 2018), test externally on MILK10k and PAD-UFES-20, test fairness on DDI, and always de-duplicate and split by lesion.

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